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United States Patent	12403612
Kind Code	B2
Date of Patent	September 02, 2025
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Robot-mounted lamp assembly for curing paint

Abstract

A robotic system is provided. The robot system includes a robot arm, a lamp assembly, and a control module. The robot arm includes a first end and a second end that opposes the first end. The lamp assembly is disposed at the second end of the robot arm and includes at least one ultraviolet (UV) light that is configured to cure paint on a panel of a vehicle. The control module is configured to actuate the robot arm and position the lamp assembly relative to the panel of the vehicle.

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Appl. No.: **18/095423**

Filed: **January 10, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240227198 A1	Jul. 11, 2024

Publication Classification

Int. Cl.: **B25J11/00** (20060101); **B05D3/06** (20060101); **B25J9/16** (20060101); **B25J15/00** (20060101)

U.S. Cl.:

CPC	**B25J11/0075** (20130101); **B05D3/067** (20130101); **B25J9/1684** (20130101); **B25J15/0019** (20130101); **B25J15/0066** (20130101);

Field of Classification Search

CPC:	B25J (11/0075); B25J (9/1684); B25J (15/0019); B25J (15/0066); B25J (9/00); B25J (9/16); B25J (9/0096); B25J (9/02); B25J (9/023); B05D (3/067)

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Background/Summary

INTRODUCTION

(1) The information provided in this section is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

(2) The present disclosure relates to painting, and more particularly to a robot-mounted lamp assembly for curing paint.

(3) During manufacture of a vehicle, paint may be applied to panels of the vehicle. More specifically, paint may be applied to an exterior surface and an interior surface of the panel. Examples of such panels include a door frame, a fuel door, and a liftgate of the vehicle. Various types of paint may be used. One type of paint is curable paint.

SUMMARY

(4) An example robotic system includes a robot arm, a lamp assembly, and a control module. The robot arm includes a first end and a second end that opposes the first end. The lamp assembly is disposed at the second end of the robot arm and includes at least one ultraviolet (UV) light that is

configured to cure paint on a panel of a vehicle. The control module is configured to actuate the robot arm and position the lamp assembly relative to the panel of the vehicle, and selectively turn on and off the at least one UV light.

(5) In one example, the at least one UV light includes at least one UV light emitting diode (LED) light.

(6) In one example, the at least one UV light includes a plurality of UV lights.

(7) In one example, each of the plurality of UV lights have the same light output wavelength.

(8) In one example, the plurality of UV lights includes at least two of a UV-A light, a UV-B light, and a UV-C light.

(9) In one example, an end effector is attached to the lamp assembly and operable to move a panel of the vehicle between a first position and a second position.

(10) In one example, the end effector is movable between an extended position and a retracted position.

(11) In one example, the end effector is disposed beside the at least one UV light.

(12) In one example, the at least one UV light includes a plurality of UV lights and the end effector is disposed radially inwardly of the plurality of UV lights.

(13) In one example, the lamp assembly includes a paint applicator.

(14) In one example, the at least one UV light includes a plurality of UV lights and the paint applicator is disposed radially inwardly of the plurality of UV lights.

(15) An example robotic system includes a robot arm, a paint applicator, a rail, a lamp assembly, and a control module. The robot arm includes a first end and a second end that opposes the first end. The paint applicator is disposed at the second end of the robot arm and is configured to output paint onto a panel of a vehicle. The rail is attached to the paint applicator and extends away from the paint applicator. The lamp assembly is configured to slide along the rail and includes at least one UV light operable to cure paint on the panel of the vehicle. The control module is configured to actuate the robot arm and position the paint applicator relative to the panel of the vehicle and position the lamp assembly relative to the panel of the vehicle.

(16) In one example, the control module is configured to slide the lamp assembly along the rail.

(17) In one example, the control module is configured to pivot the lamp assembly relative to the rail.

(18) In one example, the control module is configured to selectively move the lamp assembly toward the rail and away from the rail.

(19) In one example, the lamp assembly includes a distance sensor configured to measure a distance between the lamp assembly and the panel of the vehicle.

(20) In one example, the control module is configured to move the lamp assembly one of toward the rail and away from the rail based on the distance measured by the distance sensor.

(21) In one example, the lamp assembly includes a lamp cover disposed annularly around the at least one UV light.

(22) In one example, the at least one UV light includes a plurality of UV lights having a same wavelength.

(23) In one example, the at least one UV light includes a plurality of UV lights having more than one different wavelength.

(24) Further areas of applicability of the present disclosure will become apparent from the detailed description, the claims and the drawings. The detailed description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure will become more fully understood from the detailed description and the accompanying drawings, wherein:

(2) FIG. **1** is a functional block diagram of an example robotic system and an example vehicle;

(3) FIG. **2** is a side view of an example robot and an example lamp assembly;

(4) FIG. **3** is a bottom view of UV lights of the example lamp assembly;

(5) FIG. **4** is a bottom view of UV lights of another example lamp assembly;

(6) FIG. **5** is a side view of an example lamp assembly and an example end effector in an extended position;

(7) FIG. **6** is a side view of the example lamp assembly shown in FIG. **5** and the example end effector in a retracted position;

(8) FIG. **7** is a side view of an example lamp assembly and an example end effector in an extended position;

(9) FIG. **8** is a side view of the example lamp assembly shown in FIG. **7** and the example end effector in a retracted position;

(10) FIG. **9** is a bottom view of the example lamp assembly shown in FIGS. **7-8**;

(11) FIG. **10** is a side view of an example lamp assembly and an example end effector in an extended position;

(12) FIG. **11** is a side view of the example lamp assembly shown in FIG. **10** and the example end effector in a retracted position;

(13) FIG. **12** is a bottom view of the example lamp assembly shown in FIGS. **10-11**;

(14) FIG. **13** is a side view of an example robotic system and an example panel of the vehicle; and

(15) FIG. **14** is a side view of an example lamp assembly and an example paint applicator.

(16) In the drawings, reference numbers may be reused to identify similar and/or identical elements.

DETAILED DESCRIPTION

(17) Panels of a vehicle may be painted and cured during a production process of the vehicle. Examples of such panels include interior panels of the vehicle and exterior panels of the vehicle. In one example, the panels may be cured using a cure oven after the panel is painted. While the panels are disposed within the cure oven, the cure oven is configured to raise a temperature of the panels to a target temperature for a target duration of time. In one example, the target temperature may be about 250 degrees Fahrenheit (° F.) and the target duration of time may be about 30 minutes. However, use of the cure oven requires consumption of energy such as natural gas or electricity, which is costly and increases a carbon footprint of the production process of the vehicle.

(18) In another example, the panels of the vehicle may be cured using an ultraviolet (UV) lamp that is stationary. The UV lamp may apply UV light onto the panel for about 2 minutes. Compared against using cure ovens, the panels may be cured in less time and less energy is consumed using the UV lamp.

(19) However, the UV lamp may not cure some portions of the panels because of a poor line of sight between the UV lamp and such portions of the panels. In one example, the poor line of sight may be created by the positioning and geometry of the panel in relation to the UV lamp. Hence, such portions of the panels may be left uncured.

(20) In light of the above, the present application involves a robotic system and a robot-mounted lamp assembly for curing paint using light. FIG. **1** is a functional block diagram of an example robotic system and example vehicle.

(21) With reference to FIG. **1**, a robotic system **100** includes a control module **102**, a robot **104** and a lamp assembly **106**. The lamp assembly **106** is mounted to the robot **104** and may include UV lights. In one example, the UV lights may include UV light emitting diode (LED) lights. The control module **102** is configured to operate the robot **104** and the lamp assembly **106**. For

example, the control module **102** is configured to move the robot **104** to a location of a selected panel **108** of a vehicle **110**. The panel **108** of the vehicle **110** may be an interior panel of the vehicle **110** or an exterior panel of the vehicle **110** or another suitable panel. The panel **108** may be a swinging panel that is movable between an open position and a closed position, such as a liftgate, door, or hood. The present application is also applicable to non-swinging (e.g., fixed) panels and non-vehicle panels.

(22) After the panel **108** of the vehicle **110** is painted using curable paint (e.g., UV curable paint), the control module **102** is configured to turn on the UV lights of the lamp assembly **106**. The control module **102** is configured to move the robot **104** such that the UV lights may reach all target areas of the selected panel **108**. The robot **104** may have a suitable number of degrees of freedom (DOF), such as 6 DOF or higher. The control module **102** controls a frequency of the UV lights, an intensity of the UV lights, and a duration of time that the UV lights will on and output light to the panel **108**.

(23) FIG. **2** is a side view of an example robot **120** and an example lamp assembly **122**. FIG. **3** is a bottom view of the example lamp assembly **122**.

(24) The robot **120** may include a base **124** and a robot arm **126**. The base **124** may be positioned on a floor surface of a vehicle manufacturing plant, or alternatively on a stand or in another suitable location. The robot arm **126** extends between a first end **128** and a second end **130** that opposes the first end **128**. The first end **128** of the robot arm **126** may be attached to the base **124**. The second end **130** of the robot arm **126** may be attached to the lamp assembly **122**.

(25) The lamp assembly **122** may include a first portion **132**, a second portion **134**, and a lamp housing **136**. The first portion **132** may be directly attached to the second end **130** of the robot arm **126**. The second portion **134** may be disposed between the first portion **132** and the lamp housing **136**.

(26) As shown in FIG. **3**, the lamp housing **136** may include UV lights **138**. The UV lights **138** may be UV light emitting diode (LED) lights. In the illustrated example, the lamp housing **136** may be formed in a cylindrical shape. However, the lamp housing may be formed in another suitable shape, such as square, rectangular, or triangular.

(27) The UV lights **138** may be arranged in a pattern. In the illustrated example, a first set of UV lights **138** may be positioned along an outer diameter of the lamp housing **136**. A second set of UV lights **138** may be positioned radially inwardly of the first set of UV lights **138**. One or more UV lights **138** may be disposed radially inwardly of the second set of UV lights **138**, such as at a center **140** of the lamp housing **136**. However, the UV lights **138** may be arranged in another suitable pattern.

(28) In the illustrated example, the UV lights **138** include 19 UV lights. However, the UV lights **138** may include another suitable number of UV lights **138**. Each of the UV lights **138** may have the same output light wavelength or frequency. For example, each of the UV lights **138** may be UV-A lights. UV-A lights may have a wavelength ranging from about 315 nanometers (nm) to about 400 nm. In another example, each of the UV lights **138** may be UV-B lights. UV-B lights may have a wavelength ranging from about 280 nm to about 315 nm. In yet another example, each of the UV lights **138** may be UV-C lights. UV-C lights may have a wavelength ranging from about 100 nm to about 280 nm. In various implementations, the UV lights **138** may include a combination of two or more of UV-A, UV-B, and UV-C lights. The different UV light wavelengths may be used to cure different types of paint. For example, a primer may be cured using UV-C lights, while another paint may be cured using UV-A and/or UV-B lights.

(29) In some configurations of the lamp housing, a lamp housing may include UV lights having different wavelengths. FIG. **4** is a bottom view of an example lamp housing **150**.

(30) The UV lights may include UV-A lights **152**, UV-B lights **154**, and UV-C lights **156**, or a combination thereof. The control module **102** is configured to selectively turn on ones or sets of UV lights **152**, **154**, **156** based on the paint applied. In the illustrated example, the lamp housing

150 includes six UV-A lights **152**, seven UV-B lights **154**, and six UV-C lights **156**. However, the lamp housing **150** may include another suitable number of UV-A, UV-B and UV-C lights **152**, **154**, **156**, and the UV-A, UV-B, and UV-C lights **152**, **154**, **156** may be arranged in another suitable pattern. In one example, the lamp housing **150** may also include infrared (IR) lights. The control module **102** is configured to selectively turn on one or more of the IR lights to cure the paint, such as when heat is required.

(31) In some configurations of the lamp assembly, a lamp assembly **160** may include an end effector **162**. The end effector **162** is movable to an extended position, to a retracted position, and to positions between the extended and retracted positions. FIG. **5** is a side view of the example lamp assembly **160** with the example end effector **162** in the extended position. FIG. **6** is a side view of the example lamp assembly **160** with the example end effector **162** in the retracted position.

(32) Like the lamp assembly **122**, the lamp assembly **160** may include a first portion **164**, a second portion **166**, and a lamp housing **168**. The end effector **162** may be attached to the first portion **164** of the lamp assembly **160** and may be positioned beside the second portion **166**. The control module **102** is configured to actuate the end effector **162** between the extended position shown in FIG. **5** and the retracted position shown in FIG. **6**.

(33) In some examples, the panel **108** may move from a closed (first) position to an open (second) position to access another panel **108** or an interior portion of the same panel **108**. While the end effector **162** is in the extended position, the control module **102** is configured to move the panel **108** from the closed position to the open position using the robot (e.g., the end effector **162**). In one example, such as for a fuel door (a panel), the end effector **162** may be pressed against the panel **108** to move the panel **108** to the open position. In another example, such as for a liftgate (a panel), the end effector **162** may operate a control portion (e.g., a button or switch) of the panel **108** to move the panel **108** to the open position. The control module **102** is configured to operate the end effector **162** to move the panel **108** from the open position to the closed position.

(34) The end effector **162** may include a rod **170** and a contact device **172** attached to the rod **170**. The rod **170** may extend between a third end **174** and a fourth end **176** that opposes the third end **174**. The contact device **172** may be attached to the fourth end **176** of the rod **170**. In the illustrated example, the contact device **172** is shown as a suction cup. However, the contact device **172** may be another suitable type of contact device used to actuate and/or physically contact the panel **108** of the vehicle **110**, such as a hook. When the end effector **162** is in the extended position, the rod **170** may extend from the first portion **164** such that both the third and fourth ends **174**, **176** of the rod **170** are disposed outside the first portion **164**. When the end effector **162** is in the retracted position, the rod **170** may be substantially retracted into the first portion **164** such that the third end **174** of the rod **170** is disposed within the first portion **164** and the contact device **172** is disposed outside the first portion **164**.

(35) In some configurations of the lamp assembly having the end effector, an end effector may be attached to (e.g., within) a lamp housing of a lamp assembly. FIG. **7** is a side view of an example lamp assembly **180** with an example end effector **182** in an extended position. FIG. **8** is a side view of the example lamp assembly **180** shown in FIG. **7** with the example end effector **182** in a retracted position. FIG. **9** is a bottom view of the example lamp assembly **180** shown in FIGS. **7-8**.

(36) Like the lamp assemblies **122**, **160**, the lamp assembly **180** may include a first portion **184**, a second portion **186**, and a lamp housing **188** having UV lights **189**. The end effector **182** may be attached to the lamp housing **188**. The control module **102** is configured to move the end effector **182** between the extended position shown in FIG. **7** and the retracted position shown in FIG. **8**, and configured to move the panel **108** between the open and closed positions using the end effector **182**.

(37) Like the end effector **162**, the end effector **182** may include a rod **190** and a contact device **192** attached to the rod **190**. The rod **190** may extend between a third end **194** and a fourth end **196** that

opposes the third end **194**. When the end effector **182** is in the extended position, the rod **190** may extend from the lamp housing **188** such that both the third and fourth ends **194**, **196** of the rod **190** are disposed outside the lamp housing **188**. When the end effector **182** is in the retracted position, the rod **190** may be retracted into the lamp assembly **180** such that the third end **194** of the rod **190** is disposed within the second portion **186** and the contact device **192** is disposed within the lamp housing **188**.

(38) In the illustrated example shown in FIG. **9**, the end effector **182** may be positioned at a center **200** of the lamp housing **188** and the UV lights **189** may be disposed annularly around the end effector **182** and radially outwardly of the end effector **182**. However, the end effector **182** may be positioned in another suitable location in or on the lamp housing **188**.

(39) In some configurations of the lamp assembly, a lamp assembly may include a paint applicator. FIG. **10** is a side view of an example lamp assembly **210** with an example end effector **212** in an extended position and an example paint applicator **214**. FIG. **11** is a side view of the example lamp assembly **210** shown in FIG. **10** with the example end effector **212** in a retracted position and the example paint applicator **214**. FIG. **12** is a bottom view of the example lamp assembly **210** shown in FIGS. **10-11**.

(40) Like the lamp assemblies **122**, **160**, **180**, the lamp assembly **210** may include a first portion **216**, a second portion **218**, and a lamp housing **220** having UV lights **222**. The end effector **212** may be attached to the first portion **216** of the lamp assembly **210** and may be disposed beside the second portion **218**. The control module **102** is configured to move the end effector **212** between the extended position shown in FIG. **10** and the retracted position shown in FIG. **11** and configured to move the panel **108** between the open and closed positions using the end effector **212**.

(41) Like the end effectors **162**, **182**, the end effector **212** may include a rod **224** and a contact device **226** attached to the rod **224**. The rod **224** may extend between a third end **228** and a fourth end **230** that opposes the third end **228**. When the end effector **212** is in the extended position, the rod **224** may extend from the first portion **216** such that both the third and fourth ends **228**, **230** of the rod **224** are disposed outside the first portion **216**. When the end effector **212** is in the retracted position, the rod **224** may be substantially retracted into the first portion **216** such that the third end **228** of the rod **224** is disposed within the first portion **216** and the contact device **226** is disposed outside the first portion **216**.

(42) The paint applicator **214** may be a high transfer efficiency paint applicator or another suitable type of paint applicator. The paint applicator **214** may be disposed within the lamp assembly **210**. More specifically, the paint applicator **214** may extend through the first portion **216**, second portion **218**, and the lamp housing **220**. In the illustrated example shown in FIG. **12**, the paint applicator **214** may be positioned at a center **232** of the lamp housing **220** and the UV lights **222** may be disposed annularly around the end effector **212** and radially outward of the paint applicator **214**. However, the paint applicator **214** may be positioned in another suitable location in or on the lamp housing **220**.

(43) The control module **102** is configured to actuate the paint applicator **214** to output paint onto the panel **108**. After painting, the control module **102** is configured to turn on the UV lights **222** to cure the applied paint.

(44) In some configurations of the robotic system, a paint applicator may be attached to a robot and a lamp assembly may be slidably attached to the paint applicator. FIG. **13** is a side view of an example robotic system **240** and the panel **108**. FIG. **14** is a side view of an example paint applicator **242**, a rail assembly **244**, and an example lamp assembly **246** of the robotic system shown in FIG. **13**.

(45) The robotic system **240** may include the robot **120**, the paint applicator **242**, the rail assembly **244**, and the lamp assembly **246**. The paint applicator **242** may include a housing **248** and a nozzle **250** attached to the housing **248**. The housing **248** is directly attached to the second end **130** of the robot arm **126**. The nozzle **250** may be configured to output (e.g., spray) curable paint. In the

illustrated example, the nozzle **250** may be formed in a cone shape. However, the nozzle **250** may be formed in another suitable shape. The paint applicator **242** may be a high transfer efficiency paint applicator or another suitable type of paint applicator.

(46) The rail assembly **244** is attached to the paint applicator **242** using a sleeve **252** or another suitable type of attachment/connection device. The rail assembly **244** extends away from the housing **248** of the paint applicator **242**. The rail assembly **244** may include a rail **254** and a slider **256**. The rail **254** may extend between a fifth end **258** and a sixth end **260** that opposes the fifth end **258**. The fifth end **258** of the rail **254** may be attached to the paint applicator **242**. The sixth end **260** of the rail **254** may be positioned longitudinally away from the paint applicator **242**. Thus, the rail **254** may extend in a substantially longitudinal direction. The slider **256** may be coupled to the rail **254** and may be slidable along the rail **254** between the fifth and sixth ends **258**, **260** of the rail **254**.

(47) The lamp assembly **246** may be attached to the slider **256** of the rail assembly **244** such that the lamp assembly **246** is positioned adjacent to the paint applicator **242**. The control module **102** is configured to move the lamp assembly **246** along the rail **254**.

(48) The lamp assembly **246** may include a connector **262**, a lamp housing **264**, and a lamp shield **266**. The connector **262** attaches the slider **256** of the rail assembly **244** to the lamp housing **264**. The lamp housing **264** may be the same or substantially similar to the lamp housings **136**, **150**. Specifically, the lamp housing **264** includes UV lights. The lamp housing **264** may additionally include a distance sensor **268**. The distance sensor **268** is configured to measure a distance between the UV lights of the lamp housing **264** and the panel **108** of the vehicle **110**.

(49) The lamp shield **266** may include a lamp panel **270** and a lamp cover **272**. The lamp panel **270** may be disposed on top of the lamp housing **264** and may extend outwardly from the lamp housing **264**. In the illustrated example, the lamp panel **270** is formed into a circular shape. However, the lamp panel **270** may be formed in another suitable shape. The lamp cover **272** may extend from a perimeter of the lamp panel **270** such that the lamp cover **272** is disposed annularly around the lamp housing **264**. The lamp cover **272** may be made of a yellow-dyed cellulose triacetate film, or any other suitable material.

(50) The control module **102** may simultaneously activate the paint applicator **242** to output (e.g., spray) paint and turn on the UV lights of the lamp housing **264** to cure paint. For example, the paint applicator **242** may apply curable paint to a first portion **274** of the panel **108** while the UV light cures a second portion **276** of the panel **108** that has been already painted. The lamp cover **272** is positioned to restrict exposure of the UV light from the lamp assembly **246** to the first portion **274** of the panel **108** that is being painted to avoid premature curing.

(51) The control module **102** is configured to move the lamp assembly **246** along the slider **256**. In some examples, the lamp assembly **246** alternatively or additionally may be moved along the slider **256** manually by an operator. The control module **102** determines an off-set distance to move the lamp assembly **246** along the slider **256** based on a duration of time for the curable paint to settle before starting the curing process.

(52) The control module **102** is configured to move the lamp assembly **246** relative to the slider **256**. In the illustrated example, the connector **262** is configured to move in a substantially vertical direction, thereby moving the lamp housing **264** in a corresponding direction vertically upwardly and downwardly. Additionally, the lamp housing **264** may be configured to rock (e.g., pivot) relative to the connector **262**. The lamp housing **264** may rock in another suitable direction. The control module **102** determines a movement of the lamp assembly **246** based on the distance from the distance sensor **268**. More specifically, the control module **102** adjusts an output intensity of the UV lights and the distance from the distance sensor **268** in order to achieve a target intensity at the panel **108**.

(53) The foregoing description is merely illustrative in nature and is in no way intended to limit the disclosure, its application, or uses. The broad teachings of the disclosure can be implemented in a

variety of forms. Therefore, while this disclosure includes particular examples, the true scope of the disclosure should not be so limited since other modifications will become apparent upon a study of the drawings, the specification, and the following claims. It should be understood that one or more steps within a method may be executed in different order (or concurrently) without altering the principles of the present disclosure. Further, although each of the embodiments is described above as having certain features, any one or more of those features described with respect to any embodiment of the disclosure can be implemented in and/or combined with features of any of the other embodiments, even if that combination is not explicitly described. In other words, the described embodiments are not mutually exclusive, and permutations of one or more embodiments with one another remain within the scope of this disclosure.

(54) Spatial and functional relationships between elements (for example, between modules, circuit elements, semiconductor layers, etc.) are described using various terms, including "connected," "engaged," "coupled," "adjacent," "next to," "on top of," "above," "below," and "disposed." Unless explicitly described as being "direct," when a relationship between first and second elements is described in the above disclosure, that relationship can be a direct relationship where no other intervening elements are present between the first and second elements, but can also be an indirect relationship where one or more intervening elements are present (either spatially or functionally) between the first and second elements. As used herein, the phrase at least one of A, B, and C should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR, and should not be construed to mean "at least one of A, at least one of B, and at least one of C."

(55) In the figures, the direction of an arrow, as indicated by the arrowhead, generally demonstrates the flow of information (such as data or instructions) that is of interest to the illustration. For example, when element A and element B exchange a variety of information but information transmitted from element A to element B is relevant to the illustration, the arrow may point from element A to element B. This unidirectional arrow does not imply that no other information is transmitted from element B to element A. Further, for information sent from element A to element B, element B may send requests for, or receipt acknowledgements of, the information to element A.

(56) In this application, including the definitions below, the term "module" or the term "controller" may be replaced with the term "circuit." The term "module" may refer to, be part of, or include: an Application Specific Integrated Circuit (ASIC); a digital, analog, or mixed analog/digital discrete circuit; a digital, analog, or mixed analog/digital integrated circuit; a combinational logic circuit; a field programmable gate array (FPGA); a processor circuit (shared, dedicated, or group) that executes code; a memory circuit (shared, dedicated, or group) that stores code executed by the processor circuit; other suitable hardware components that provide the described functionality; or a combination of some or all of the above, such as in a system-on-chip.

(57) The module may include one or more interface circuits. In some examples, the interface circuits may include wired or wireless interfaces that are connected to a local area network (LAN), the Internet, a wide area network (WAN), or combinations thereof. The functionality of any given module of the present disclosure may be distributed among multiple modules that are connected via interface circuits. For example, multiple modules may allow load balancing. In a further example, a server (also known as remote, or cloud) module may accomplish some functionality on behalf of a client module.

(58) The apparatuses and methods described in this application may be partially or fully implemented by a special purpose computer created by configuring a general purpose computer to execute one or more particular functions embodied in computer programs. The functional blocks, flowchart components, and other elements described above serve as software specifications, which can be translated into the computer programs by the routine work of a skilled technician or programmer.

(59) The computer programs include processor-executable instructions that are stored on at least one non-transitory, tangible computer-readable medium. The computer programs may also include

or rely on stored data. The computer programs may encompass a basic input/output system (BIOS) that interacts with hardware of the special purpose computer, device drivers that interact with particular devices of the special purpose computer, one or more operating systems, user applications, background services, background applications, etc.

(60) The computer programs may include: (i) descriptive text to be parsed, such as HTML (hypertext markup language), XML (extensible markup language), or JSON (JavaScript Object Notation) (ii) assembly code, (iii) object code generated from source code by a compiler, (iv) source code for execution by an interpreter, (v) source code for compilation and execution by a just-in-time compiler, etc. As examples only, source code may be written using syntax from languages including C, C++, C#, Objective-C, Swift, Haskell, Go, SQL, R, Lisp, Java®, Fortran, Perl, Pascal, Curl, OCaml, Javascript®, HTML5 (Hypertext Markup Language 5.sup.th revision), Ada, ASP (Active Server Pages), PHP (PHP: Hypertext Preprocessor), Scala, Eiffel, Smalltalk, Erlang, Ruby, Flash®, Visual Basic®, Lua, MATLAB, SIMULINK, and Python®.

Claims

1. A robotic system comprising: a robot arm including a first end and a second end that opposes the first end; a paint applicator disposed at the second end of the robot arm and configured to output paint onto a panel of a vehicle; a rail attached to the paint applicator and extending away from the paint applicator; and a lamp assembly configured to slide along the rail and that includes at least one UV light operable to cure paint on the panel of the vehicle; and a control module configured to actuate the robot arm and position the paint applicator relative to the panel of the vehicle and position the lamp assembly relative to the panel of the vehicle.
2. The robotic system of claim 1, wherein the control module is configured to slide the lamp assembly along the rail.
3. The robotic system of claim 1, wherein the control module is configured to pivot the lamp assembly relative to the rail.
4. The robotic system of claim 1, wherein the control module is configured to selectively move the lamp assembly toward the rail and away from the rail.
5. The robotic system of claim 1, wherein the lamp assembly includes a distance sensor configured to measure a distance between the lamp assembly and the panel of the vehicle.
6. The robotic system of claim 5, wherein the control module is configured to move the lamp assembly one of toward the rail and away from the rail based on the distance measured by the distance sensor.
7. The robotic system of claim 1, wherein the lamp assembly includes a lamp cover disposed annularly around the at least one UV light.
8. The robotic system of claim 1, wherein the at least one UV light includes a plurality of UV lights having a same wavelength.
9. The robotic system of claim 1, wherein the at least one UV light includes a plurality of UV lights having more than one different wavelength.